



Electrochemical performance of novel electrolyte solutions based on organoboron magnesium salts

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ABSTRACT

A novel boron based electrolyte system based on reaction of tri(3,5-dimethylphenyl)borane (Mes_3B) and PhMgCl in THF solution is proposed for the potential application in rechargeable magnesium battery. The optimized composition of $\text{Mes}_3\text{B}-(\text{PhMgCl})_2/\text{THF}$ electrolyte solution presents high ionic conductivity (ca. $2 \times 10^{-3} \text{ S/cm}$ in 0.5 M solution) at room temperature, excellent Mg deposition reversibility as well as the highest anodic potential to date (3.5 V vs. Mg RE). The initial results indicate that the high-performance of the $\text{Mes}_3\text{B}-(\text{PhMgCl})_2$ electrolyte might be attributed to non-covalent interactions arising from the excess amount of Grignard reagent PhMgCl . This study opens the door to the development of boron based electrolyte chemistry for high-energy rechargeable Mg batteries.

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1. Introduction

Rechargeable Mg batteries have gained more and more attention as next-generation energy storage devices because they have potential advantages (safety, low cost and long life) in heavy load applications [1–3]. Nevertheless, the development of Mg electrochemistry suffers from the fatal problems due to the lack of suitable cathode materials and high-performance electrolyte systems [4]. Finding a suitable electrolyte with a wide electrochemical window, high Mg deposition reversibility and ion conductivity is one of the key issues in developing high-energy rechargeable Mg batteries.

It is well known that Mg electrodes display highly reversible behavior in organometallic Mg salt solutions such as Grignard salts in ethereal solutions (RMgX , R = alkyl, aryl groups; $X = \text{Cl}, \text{Br}$) [5–7]. However, they cannot be used in Mg batteries because of their narrow electrochemical windows [8]. The landmark progress was achieved by Aurbach et al., who developed a family of electrolyte solutions based on Mg organohaloaluminate salts $\text{Mg}(\text{AlCl}_4 - n\text{R}_n\text{R}'_n)_2$ (R, R' = alkyl or aryl groups, $n' + n'' = n$) in ethereal solvents. The optimized composition of THF solution of $\text{Mg}(\text{AlCl}_2\text{BuEt})_2$ complex and $\text{AlCl}_3-(\text{PhMgCl})_2$ complex exhibited improved anti-oxidation capability and excellent reversible Mg deposition/dissolution characteristics, which greatly promoted the development of rechargeable Mg batteries [3,9].

Organic boron Mg complex salts (OBMCs) would be also promising electrolyte candidates for rechargeable Mg batteries. However,

there have not been any corresponding reports apart from the work of Gregory et al. in 1990 [1], who reported that a $\text{Mg}(\text{BPh}_2\text{Bu}_2)_2$ solution (Ph and Bu represent phenyl and butyl groups respectively) used as electrolyte showed efficient reversible Mg deposition–dissolution process, and an improved anodic stability compared with some common Grignard salt solutions (1.9 V vs. 1.5 V). This result disclosed for the first time that boron-based organic electrolytes could be feasibly applied in rechargeable Mg batteries. Aurbach et al. also investigated THF solutions of Bu_2Mg as Lewis base, together with different boron based Lewis acids (BPh_2Cl , BPhCl_2 , $\text{B}[(\text{CH}_3)_2\text{N}]$, BPh_3 , BET_3 , BBR_3 , BCl_3 and BF_3). However, the results show these boron based electrolyte have low Mg cycling efficiency and lower anodic stabilities (about 1.5 V) [9].

Here we propose a new electrolyte system based on reaction products of tri(3,5-dimethylphenyl)borane (Mes_3B) and PhMgCl in THF solvent. The ion conductivity, reversibility of Mg deposition and dissolution and anodic stability of the novel boron based electrolyte system has been investigated with regard to the potential application in rechargeable magnesium battery.

2. Experiment

Tri(3,5-dimethylphenyl)borane compounds (Mes_3B) was synthesized using a modified organometallic route [10] involving the direct reaction of organic halide, Mg, and $\text{BF}_3 \cdot \text{OEt}_2$. 2,6-diisopropylphenylmagnesium bromide ($\text{Ph}_{2,6\text{-diipr}}\text{MgBr}$) was synthesized according to reference [11]. The boron based organic Mg electrolytes were synthesized by the reaction of Mes_3B Lewis acid and Grignard reagent Lewis bases in different concentrations and molar ratios following the synthesis procedure of $[\text{Mes}_3\text{BPh}]^-[\text{MgCl}]^+$. First, Mes_3B (0.332 g,

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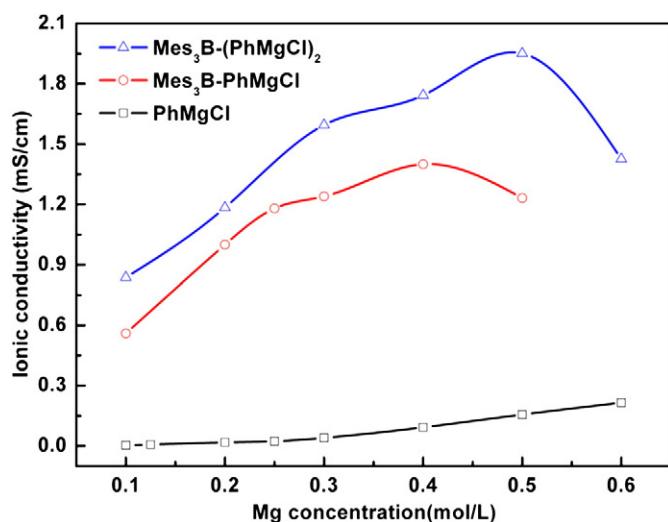


Fig. 1. The specific ion conductivities of THF solutions containing different concentration of Mg salts (calculate based on Mg) at room temperature.

1 mmol) was dissolved in highly pure THF (3 ml), and then, PhMgCl/THF solution (Aldrich, 2 M, 0.5 ml, 1 mmol) was slowly added to the Mes₃B solution. The resulting solution was stirred for several hours at room temperature. The solvent was removed under reduced pressure, and the product was afforded as solid powder in essentially quantitative yield. ¹H NMR (*d*₈-THF, 400 MHz): δ 2.04 (s, 18H, methyl protons), 6.31 (s, 3H, aromatic protons), 6.61–6.89, 7.20–7.25 (m, 11H, aromatic protons); ¹³C NMR (*d*₈-THF, 100 MHz): δ 23.4 (methyl carbons), 122.5, 122.9, 124.6, 126.8, 130.1, 134.0, 136.5, 138.2 (aromatic carbons); ¹¹B NMR (CDCl₃, 128.4 MHz): δ –26.2.

All the electrochemical measurements and XRD and SEM measurements were carried out according to the previously reported literature procedure [7].

3. Results and discussion

By varying the molar ratio of acid to base as well as the reaction time, the optimum electrochemical performance of the electrolyte is obtained when the mole ratio of Mes₃B:PhMgCl is 1:2 and the reaction time is 10 h. Fig. 1 shows the specific conductivity of different concentration solutions containing the reaction products of Mes₃B with PhMgCl in THF at various reactant ratios. In general, higher conductivity is expected for more concentrated salt solutions. This is true in a low concentration range, but fails in a highly concentrated

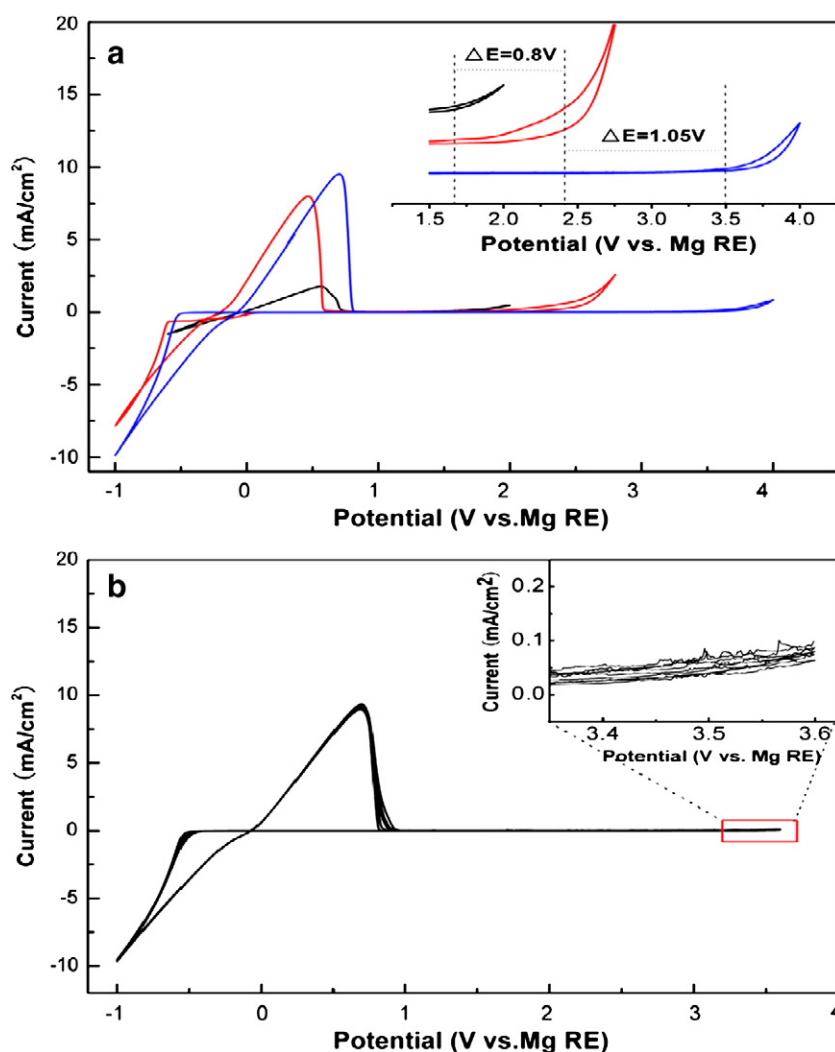


Fig. 2. Typical steady-state voltammograms of Pt electrodes in different electrolytes. (a) One molar PhMgCl (black), 0.4 M Mes₃B–PhMgCl (red) and 0.5 M Mes₃B–(PhMgCl)₂ (blue); the inset is the enlargement of 1.5–4 V region of Fig. 2a. (b) Five cycles in 0.5 M Mes₃B–(PhMgCl)₂/THF solution; the inset is the anodic region of CVs. The scan rate was 50 mV/S. All the electrolyte concentrations were calculated based on Mg content.

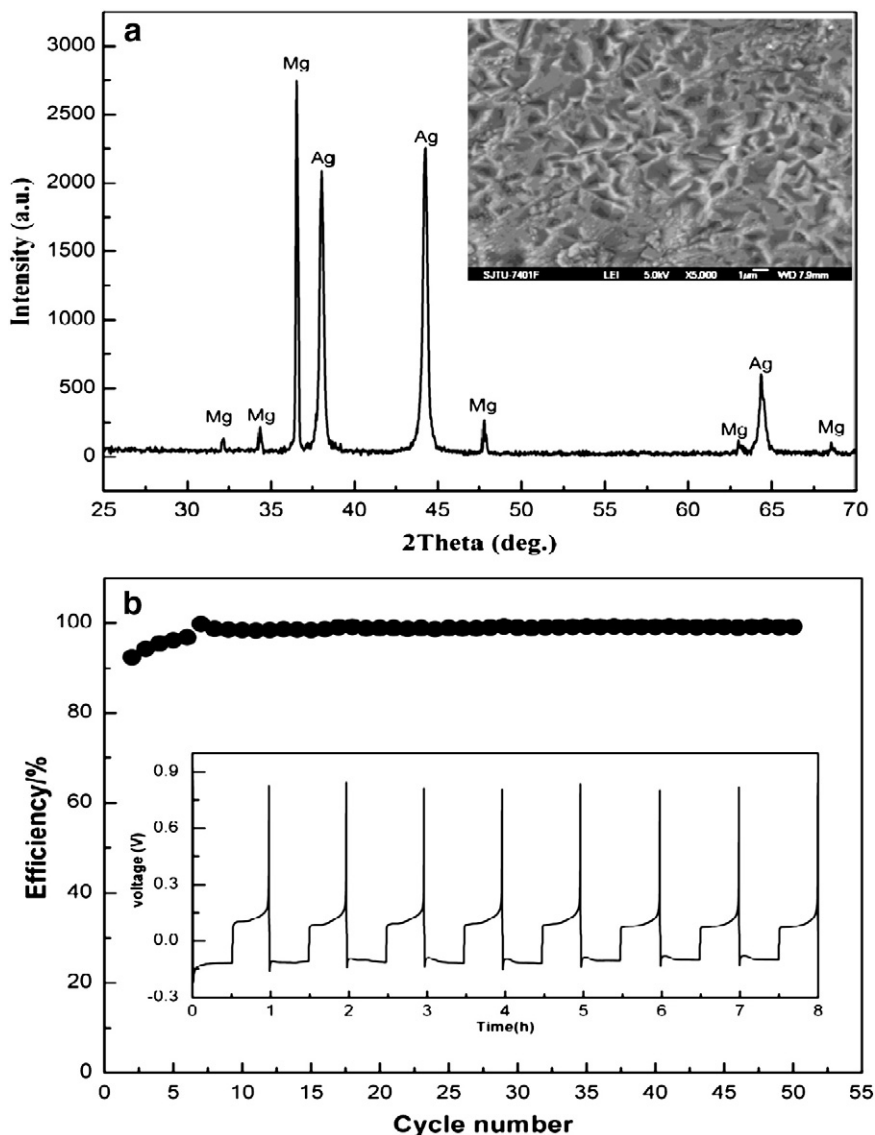


Fig. 3. (a) XRD pattern of the electrodeposited Mg on Ag substrate in 0.5 M Mes₃B–(PhMgCl)₂/THF solution and its SEM image. The charge amount is 1.8 C/cm². (b) The cycling efficiencies and the first eight chronopotentiograms of Mg deposition and dissolution on Ag in 0.5 M Mes₃B–(PhMgCl)₂/THF solution.

electrolyte solution due to the formation of ion pair and the enhanced viscosity. As shown in Fig. 1, the room temperature ionic conductivities of Mes₃B–PhMgCl/THF and Mes₃B–(PhMgCl)₂/THF solutions rise monotonically with the enhanced salt concentration and reach their peak values. In particular, it is noticed that the ion conductivity of Mes₃B–(PhMgCl)₂ electrolyte solution is much higher than that of Mes₃B–PhMgCl, which demonstrates the highest value of about 2×10^{-3} S/cm in 0.5 M solution at room temperature, comparable to the electrolytes used in Li batteries.

The Mg deposition reversibility and the anodic stability of the OBMCS electrolyte were measured by cyclic voltammetry (CV) with the Pt working electrode, as shown in Fig. 2a; reversible Mg deposition–dissolution processes could be achieved in the OBMCS electrolytes. The electrochemical window of Mes₃B–PhMgCl solution exhibits about 2.5 V vs. Mg RE, much higher than that of pure PhMgCl/THF electrolyte (1.7 V) and reported Mg(BPh₂Bu)₂ solution (1.9 V) [1], whose electrochemical performance was fully compared to Aubach's first generation electrolyte [9]. More interestingly, the Mes₃B–(PhMgCl)₂ solution not only delivers the highest reversible current peaks for Mg deposition and dissolution, about 7 times larger than those of pure PhMgCl/THF solution, but also exhibits a wide electrochemical window till ca. 3.5 V vs. Mg RE, which is 1 V

more positive than that of the Mes₃B–PhMgCl solution and wider than any reported in the literature to date [12–14]. The multi-cycle CVs from –1 V to 3.6 V (vs. Mg RE) in Fig. 2b further demonstrates the excellent electrochemical reversibility of the Mes₃B–(PhMgCl)₂ electrolyte system. The inset in Fig. 2b shows enlarged anodic region near end of the CVs, which is characterized by a very small change of the current density.

The deposit layers from Mes₃B–(PhMgCl)₂ electrolyte solution were characterized by XRD and SEM measurements (Fig. 3a). The typical peaks at 32.19, 34.39, 36.62, 47.8, 57.37, 63.06, 67.3 and 68.6 (2θ) in XRD pattern are in accordance with hexagonal close-packed structure of Mg (JCPDS-350821), and no peaks from other phases can be detected, thus indicating that the deposited Mg is of high purity. SEM image indicates the uniform morphology of Mg deposition, which is very important for the practical application of a secondary battery due to avoiding metallic dendrite formation, which often occurs on lithium electrodes [15]. The critical parameter of coulombic efficiency of Mg deposition/dissolution processes was evaluated by galvanostatic cycling experiments (Fig. 3b), calculated according to the ratio of the charge amount of Mg dissolution to that of Mg deposition (Fig. 3b inset). The coulombic efficiency of deposition/dissolution cycle quickly approaches to 100% from 92% of the first

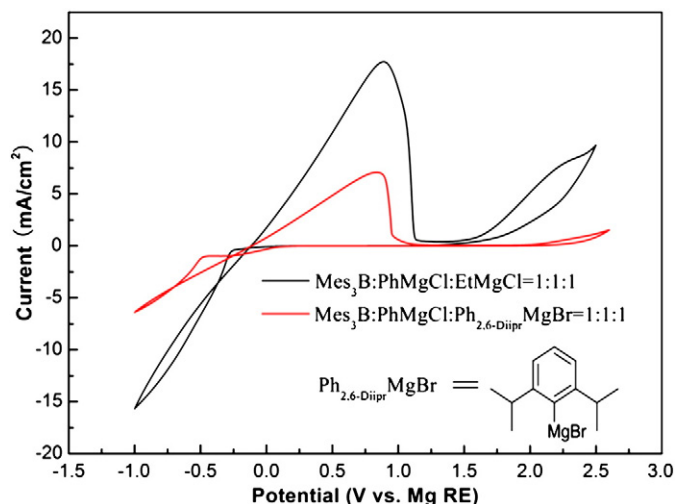


Fig. 4. Steady-state cyclic voltammograms of Pt electrode in different electrolyte solutions: (black) $\text{Mes}_3\text{B}:\text{PhMgCl}:\text{EtMgCl} = 1:1:1$, (red) $\text{Mes}_3\text{B}:\text{PhMgCl}:\text{Ph}_{2,6\text{-diipr}}\text{MgBr} = 1:1:1$, scanning rate 50 mV/S.

cycle after 6 cycles. The high ionic conductivity, outstanding electrochemical reversibility and anodic stability make $\text{Mes}_3\text{B}-(\text{PhMgCl})_2$ complex electrolyte highly promising for use in high-energy rechargeable Mg batteries.

The unexpected high electrochemical performance of $\text{Mes}_3\text{B}-(\text{PhMgCl})_2$ electrolyte solution is fascinating. Combined with the ^{11}B , ^1H and ^{13}C NMR spectra analyses, it can be concluded that the Mes_3B Lewis acid can react with 1 Eq PhMgCl to form the stable $[\text{Mes}_3\text{BPh}]^-[\text{MgCl}]^+$ compound. It was assumed that the most oxidation vulnerable species in the electrolyte governs the solution anodic stability [16]. Reasonably, the $\text{Mes}_3\text{B}-(\text{PhMgCl})_2$ electrolyte is a mixed solution of $[\text{Mes}_3\text{BPh}]^-[\text{MgCl}]^+$ compound and another one equivalent PhMgCl , and its anodic stability should be very low if determined by PhMgCl . However, it is surprising that the $\text{Mes}_3\text{B}-(\text{PhMgCl})_2$ electrolyte shows the highest anodic stability. Hence, we speculated that this phenomenon relates to the two equivalents of PhMgCl in $\text{Mes}_3\text{B}-(\text{PhMgCl})_2$ that not only provide one equivalent of Lewis base to form stable cation $[\text{Mes}_3\text{BPh}]^+$, but also act as critical species to construct effective compositions via non-covalent interaction (such as π - π interaction [17]) for improving the electrochemical performance of the whole electrolyte system. The electrochemical performance of $[\text{Mes}_3\text{BPh}]^-[\text{MgCl}]^+$ compound coupling with non-aromatic Grignard EtMgCl or sterically-bulky aromatic 2,6-diisopropylphenylmagnesium bromide ($\text{Ph}_{2,6\text{-diipr}}\text{MgBr}$) seems to give a strong support for the above analyses. In the former case, there is no π - π weak interaction between Et group of EtMgCl and $[\text{Mes}_3\text{BPh}]^-$ anion. Also, $\text{Ph}_{2,6\text{-diipr}}$ group cannot effectively generate π - π interaction with $[\text{Mes}_3\text{BPh}]^-$ anion due to the larger steric effect. Accordingly, the lower anodic potentials were observed for these two cases at 1.5 and 2.0 V, similar to the electrochemical behaviors of pure EtMgCl and $\text{Ph}_{2,6\text{-diipr}}\text{MgBr}$ electrolyte solutions (see Fig. 4), respectively.

4. Conclusions

We have developed a facile approach to a low-cost novel boron-based organic Mg salt electrolyte system using a reaction between boron-based Lewis acid and Grignard reagent as Lewis base. The resulting $\text{Mes}_3\text{B}-(\text{PhMgCl})_2/\text{THF}$ electrolyte system presents high ionic conductivity, excellent electrochemical reversibility of Mg deposition/dissolution and the highest anodic potential up to 3.5 V to date. This made the novel boron-based electrolyte highly promising for use in high-energy rechargeable Mg batteries. The initial results indicate that the high-performance of the $\text{Mes}_3\text{B}-(\text{PhMgCl})_2$ electrolyte might be attributed to non-covalent interactions arising from the excess amount of Grignard reagent PhMgCl . However, due to the complicated ion compositions, interaction and solvation effect in $\text{Mes}_3\text{B}-(\text{PhMgCl})_2$ electrolyte solution, we cannot be sure of the exact function mechanism at this stage. Further works to understand the reason for the fantastic effect and find the suitable high potential cathode material to construct high-energy rechargeable Mg battery using the novel boron based electrolyte are in progress.

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